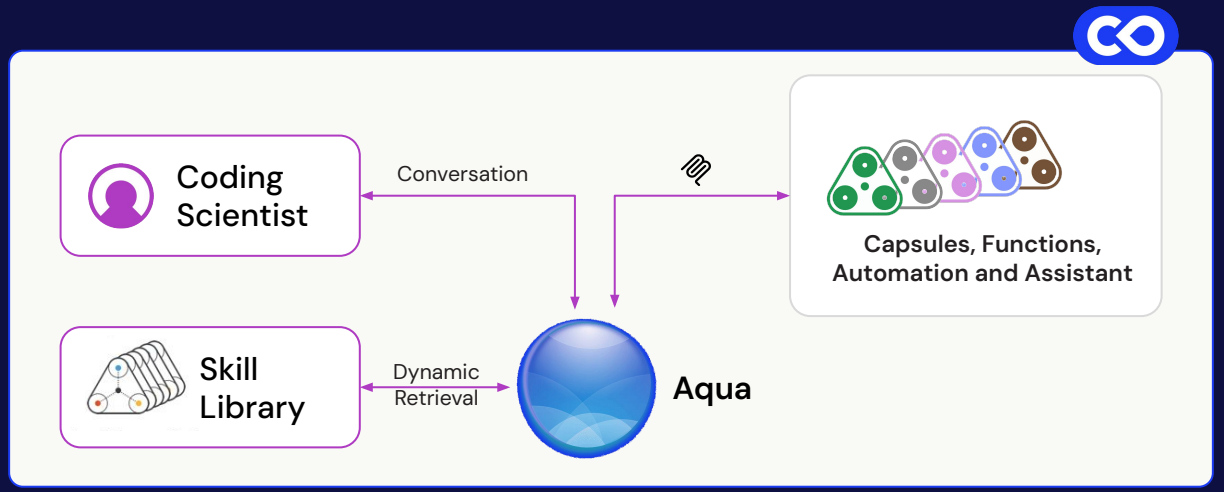


code ocean

Aqua Custom Skills for Coders and Bioinformaticians

October 2026

Welcome To Custom Skills Within Code Ocean!



Agenda

- **Skills Background:**
  - What are Skills and what can they do?
  - Types of Skills
  - Aqua behavior policies
- **Skill Fundamentals:**
  - Components
  - Best practices for skill design
  - Building, sharing and metadata
  - Enabling/disabling and invoking
- **Examples**
  - always-beep-boop
  - app-panel-construction
  - environment-build-troubleshooting
  - rna-seq-pathway-explorer-app
- **FAQ / Q&A**

Skills – What are they?

- Skills teach Aqua what it doesn't already know, extending its capabilities and customizing it for each user or group → **Fundamentally a set of instructions and when to use them**
  - Knowledge – teach Aqua something specific about your work
  - Procedure – standardize and give instructions on how to approach a problem
  - New Capabilities – create new behaviors or ways to use Aqua e.g. "Create a summarized memory at the end of each conversation we have and save it to a specific Capsule"



Types of Skills

Internal platform skills

Code Ocean-owned · not user-editable

- Govern how Aqua decides, routes and uses resources
- Routing policy, design, cloud workstation, planning, review

Internal science skills

Code Ocean-developed

- The science index plus 141 method skills
- Methods, invariants, hard stops and validated scripts
- UI display of the active science skill is planned for 4.11

Custom skills

Customer-owned · 'My Skills' (4.9)

- Created by asking Aqua, or imported from Git
- Can be shared per user
- Can override an internal science skill

How a skill gets loaded

The system prompt holds a compact catalogue of names and descriptions in the context window

→

Aqua matches the task to a skill and calls the skills tool.

→

The full SKILL.md loads. Reference files and scripts are read or installed only as needed.

→

Loaded skills stay active through context summarisation for the rest of the session.

Aqua behaviour policies

No fabrication

No invented IDs, results, effect sizes, credentials or sources. Recalled facts are labelled as unverified.

No placeholder data

If an input is missing, Aqua says so and offers candidates. It never substitutes mock data.

No silent substitution

The requested method, threshold or cohort is what runs. Alternatives only run alongside.

Inspect before reasoning

Look at the actual data straight away rather than guessing its shape or content.

Report blockers

A blocker is named as a blocker, not worked around or quietly dropped.

Validation is reported honestly

Failed checks appear in the reply. Anything that could change a result is fixed and re-run.

Reuse before building

Check existing apps, capsules and pipelines first, and follow mount and app-panel conventions.

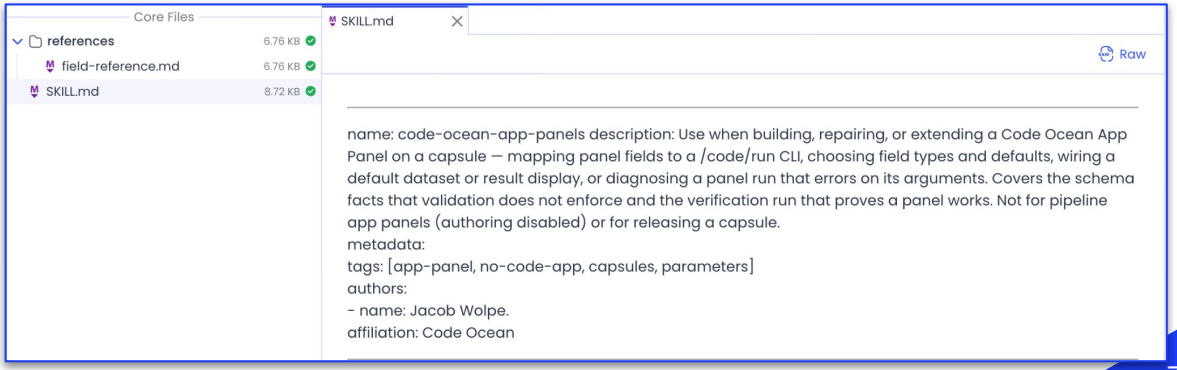
Speak the user's language

Assume a non-coder, avoid platform jargon, and offer decisions as clickable forms.

Skill Components

A Skill (or skill bundle) is a type of Capsule, consisting of:

- SKILL.md
  - Frontmatter: name (64 character limit), description (1,024 character limit), tags, author
  - Instruction body (20k character limit)
- Supporting references (optional & only used when necessary)
  - References, Scripts, Assets
- Sharing permissions
- Git repository
- Release functionality
- Enabled/disabled



Skill Building Best Practices

- One skill, one repeatable job
- Description = trigger “Use when …”
- Keep SKILL.md short → Push depth into reference files the skill links to
- Be concrete, not advisory (if, then)
- Choose the structure deliberately → Bundle or single
- Share releases
- Test on real tasks
- Keep it generalizable

To the Platform!

https://go/codeocean

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FAQ / Q&A

- **Is there a limit on number of prompts or session length?** No, aqua compacts (summarizes) messages after a given number of tokens. However, long sessions will degrade performance due to LLM amnesia
- **Are the scientific capabilities reviewed by experts in relevant fields?** The Code Ocean internal skills are derived from peer-reviewed open-source projects, and our testing has shown they enhance Aqua's performance on their specific task. They serve as a starting point for teams to make their own custom skills. Many include scripts- these are generally pulled from an expert source and are used as a template by Aqua
- **Can you customize science skills?** This feature is currently being planned, eventually, we hope users will be able to keep the ones they find useful and turn off any that aren't relevant to their work
- **Can these skills be adapted to route users to internal tools that already cover these use cases, so analyses follow standardized protocols?** Yes Custom Skills can reference specific Capsules or Data Assets
- **Does enabling all Custom Skills that have been shared with you slow Aqua down?** Aqua only sees the frontmatter to determine if it should activate a Skill. Succinct and clear frontmatter shouldn't interfere with Aqua when it is not relevant and should enable a Skill when it is. This means that the effect of the number of enabled Skills is highly variable based on their composition.